



Demographic Influences on Perceptions of Artificial Intelligence Adoption in Internal Audit: Evidence From Public Universities in Ghana

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Abstract: Artificial intelligence (AI) is increasingly transforming internal audit practices, yet empirical evidence concerning its adoption in public-sector auditing in Sub-Saharan Africa remains limited. This study examines whether age, gender, and educational level are associated with internal auditors' perceptions of AI adoption in public universities in Ghana. Drawing on the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT), a quantitative cross-sectional survey was conducted among 177 internal audit staff from six public universities. Perceptions of AI adoption were assessed across four dimensions: AI adoption and accountability, fraud detection effectiveness, implementation challenges, and strategies and enablers. Overall, a favourable orientation towards AI adoption was observed, with a composite mean score of 3.614 on a five-point scale. Multiple ordinary least squares (OLS) regression indicated that the demographic model was statistically significant, $F(3, 173) = 3.098$, $p = 0.028$, although the explanatory power was modest ($R^2 = 0.051$). Age was found to be negatively associated with perceptions of AI adoption ($B = -0.058$, $\beta = -0.185$, $p = 0.024$), indicating that more favourable perceptions were reported by younger internal auditors. Educational level showed a positive association with perceptions of AI adoption and represented the strongest predictor among the demographic variables examined ($B = 0.064$, $\beta = 0.220$, $p = 0.007$), suggesting that higher levels of academic and professional education may be associated with greater readiness for AI adoption. No statistically significant association was observed for gender ($p = 0.664$). The findings extend empirical research on technology acceptance to public-sector internal auditing in an African context and highlight the relevance of demographic heterogeneity in understanding AI adoption readiness. In particular, age-sensitive training and opportunities for continuing academic and professional development may provide appropriate mechanisms for strengthening AI-related competencies among internal audit staff in Ghanaian public universities.

Keywords: Artificial intelligence; Internal audit; Technology acceptance; Demographic factors; Ghana; Technology Acceptance Model; Unified Theory of Acceptance and Use of Technology

1. Introduction

Artificial intelligence (AI) is fundamentally reshaping the landscape of professional practice across industries, and internal auditing is no exception (Deliu & Olariu, 2024; Usul & Alpay, 2025). From AI-powered anomaly detection and continuous monitoring systems to natural language processing tools that interpret audit evidence, the potential applications of AI in audit functions are expansive and growing (Appelbaum et al., 2017; Kokina & Davenport, 2017). International bodies, including the Institute of Internal Auditors (IIA), have increasingly emphasised the imperative for internal audit professionals to develop competencies in data analytics and AI-driven methodologies.

In the Ghanaian public sector, internal audit functions operate within a structured regulatory environment anchored by the Ghana's Internal Audit Agency (IAA). Public universities, as government-funded institutions subject to legislative audit oversight, are particularly important sites for examining AI adoption in audit practice.

These institutions face intensifying demands for accountability and value-for-money assurance, and AI has the potential to substantially enhance the scope and rigour of audit coverage (Anomah, 2025). However, technology adoption in any professional setting is mediated by the attitudes, perceptions, and readiness of the individuals who must ultimately embrace it (Davis, 1989; Venkatesh et al., 2003).

A significant but underexplored dimension of this issue concerns the role of demographic characteristics in shaping how internal auditors perceive AI adoption. Practitioners differ in age, gender and educational levels, and these differences may produce meaningfully distinct orientations toward AI (Chen, 2025). Understanding these variations is essential for designing targeted interventions that promote informed and enthusiastic engagement with AI tools across diverse audit workforces. Despite the growing global literature on AI in auditing, empirical evidence from Sub-Saharan Africa and Ghana in particular remains limited. Most existing studies are anchored in Western, developed-economy contexts (Wassie & Lakatos, 2024), and their findings may not straightforwardly translate to the Ghanaian professional environment, which is characterised by distinct socio-cultural dynamics, resource constraints, and institutional configurations. This study addresses this gap by investigating how internal auditors' demographic profiles relate to their perceptions of AI adoption in Ghanaian public universities and is guided by the following research question: How do perceptions of AI adoption vary across different demographic groups among internal auditors in public universities in Ghana?

2. Theoretical and Empirical Review

2.1 Theoretical Review

2.1.1 Technology Acceptance Model

The Technology Acceptance Model (TAM), proposed by Fred Davis (1989), posits that two core cognitive beliefs, perceived usefulness (PU) and perceived ease of use (PEOU), are the determinants of a person's attitude toward using a technology, which in turn shapes behavioural intention and eventually use of it. PU refers to the degree to which a person believes that using a particular technology will enhance their job performance, while PEOU refers to the belief that interaction with the technology will be free of effort. Davis (1989) demonstrated that both constructs independently and jointly predict technology adoption across diverse organisational contexts. In the context of the present study, demographic influences on perceptions of AI adoption in internal audit functions in Ghanaian public universities, TAM provides a robust explanatory lens for understanding how auditors of different ages, genders, and educational levels form distinct perceptions of AI tools.

2.1.2 Unified Theory of Acceptance and Use of Technology

The Unified Theory of Acceptance and Use of Technology (UTAUT), developed by Venkatesh et al. (2003), identifies four core determinants of technology acceptance and use: performance expectancy, defined as the degree to which an individual believes that using the technology will help achieve gains in job performance; effort expectancy, reflecting the degree of ease associated with the use of the system; social influence, which captures how much an individual perceives that important others believe they should use the technology; and facilitating conditions, referring to the degree to which an individual believes that an organisational and technical infrastructure exists to support system use. Critically, Venkatesh et al. (2003) demonstrated that the relationships between these determinants and behavioural intention are moderated by four key demographic variables—gender, age, experience, and voluntariness of use—making UTAUT uniquely suited to studies examining demographic influences on technology adoption. By explicitly positioning age, gender, and experience as moderating variables within its predictive model, UTAUT provides both theoretical justification and a structural basis for expecting that internal auditors in Ghanaian public universities will differ in their AI adoption perceptions along demographic lines.

As illustrated in Figure 1, the theoretical framework is anchored on the TAM by Davis (1989) and the UTAUT by Venkatesh et al. (2003). The framework is structured around three interconnected columns. The left column presents the three independent variables, age, gender, and educational level, which represent the demographic characteristics of internal auditors expected to influence their AI perceptions. The middle column captures the three mediating constructs drawn from TAM and UTAUT, namely PU, PEOU, and attitude toward AI use. These constructs serve as the perceptual pathway through which demographic characteristics translate into adoption behaviour, meaning that demographics do not directly determine AI adoption but rather shape how auditors perceive AI tools, and those perceptions in turn drive adoption decisions. The right column presents the dependent variable, AI adoption outcome, operationalised through intention to adopt AI in audit, actual AI use, and AI adoption readiness. At the base of the framework, a contextual layer captures the moderating environment of Ghana's public universities, including digital infrastructure, institutional culture, and the national AI policy environment, which collectively moderate the relationships between demographics, perceptions, and adoption outcomes.

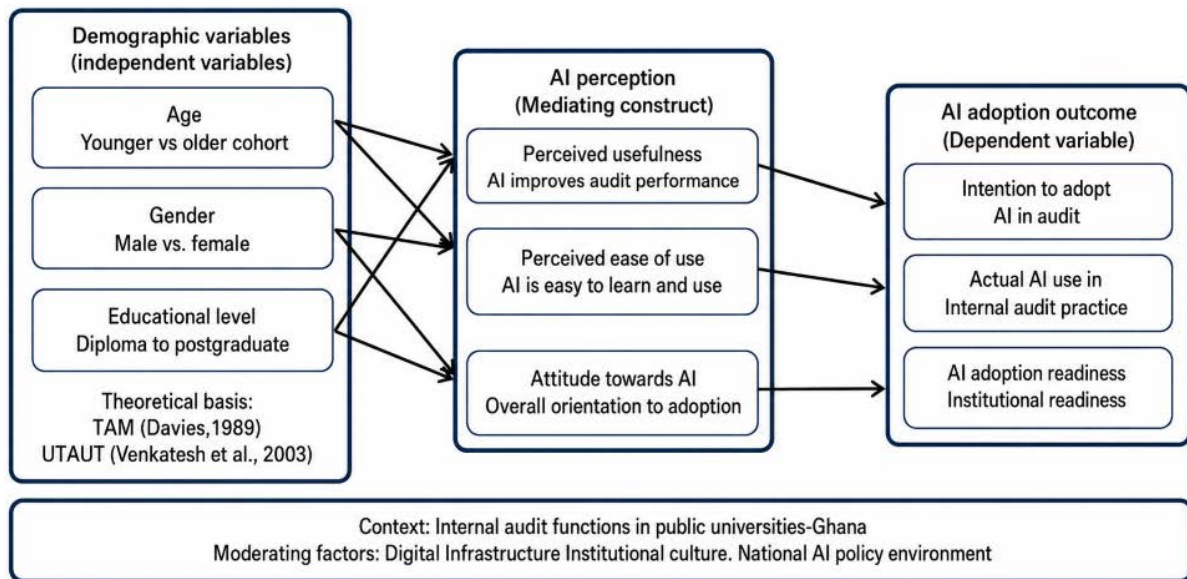


Figure 1. Conceptual framework integrating Technology Acceptance Model (TAM) and Unified Theory of Acceptance and Use of Technology (UTAUT)

Source: Adapted from Davis (1989), Venkatesh et al. (2003), and contextual Ghana artificial intelligence (AI) literature.

2.2 Empirical Review

2.2.1 Artificial intelligence in internal auditing

AI encompasses a broad spectrum of computational capabilities, including machine learning, natural language processing, robotic process automation, and predictive analytics (Ojika et al., 2024; Rane et al., 2024). In internal auditing, these capabilities translate into practical tools such as automated journal entry testing, real-time transaction monitoring, risk scoring models, and intelligent document review systems (Onyenahazi, 2025). The adoption of such tools has the potential to expand audit coverage from sampling-based to population-level testing, improve detection rates for fraud and error, and free auditors from routine procedural tasks to focus on higher-order analytical and advisory functions (Jones & Free, 2026). However, the adoption of AI in auditing is not a straightforward technological transition. It requires practitioners to revise their mental models of what audit work entails, to trust algorithmic outputs, and to acquire new technical skills alongside existing professional competencies (Celestin & Gidisu, 2023; Singh, 2026). These demands may be experienced differently across audit workforces depending on who the auditors are, their educational preparation, career trajectories, generational orientations toward technology, and positioning within professional hierarchies (Mahjoubi et al., 2026; Pérez-Calderón et al., 2025).

AI in internal auditing has expanded rapidly over the past years. Wolfe et al. (2025) revisited UTAUT specifically for the AI era, arguing that traditional adoption constructs demanded extension to capture algorithmic trust and perceived accountability in AI-assisted decision-making. Anomah (2025) explored institutional readiness and capacity for AI adoption in public audit institutions in developing countries, including Ghana, highlighting infrastructural and governance gaps that shape adoption beyond individual attitudes. Using a PRISMA-guided systematic review covering three decades of literature (1992–2024), Saifudin et al. (2025) found that research on AI and fraud detection remains concentrated in external audit, finance, and information-systems domains, with internal audit contexts still comparatively under-represented among the most-cited studies. Kontogeorgis (2025) reviewed the international regulatory landscape for AI, including the EU AI Act and IIA Standards 1100–2440, alongside survey evidence that internal audit practitioners rank improved audit effectiveness and efficiency (90%), reduced audit costs (63%), and reduced human error (63%) among the leading benefits of AI adoption, though this evidence is descriptive and not linked to auditors' individual demographic characteristics. Qatawneh (2025) provided quantitative evidence, from a sample of accounting and financial managers in Jordan, that AI-empowered accounting information systems significantly improve auditing and fraud detection, and that natural language processing significantly moderates this relationship. Jones & Free (2026) reviewed what accountants need to know about AI and machine learning, calling for further empirical work on how demographic and organisational factors jointly condition adoption outcomes. The present study differs from this recent body of work in three respects: it isolates internal auditors, rather than external auditors, financial managers, or accountants broadly, as the centre of analysis; it scrutinises individual-level adoption perceptions rather than firm-level fraud-detection outcomes or institutional frameworks; and it jointly models age, gender, and educational level as predictors within a single regression framework in a Ghanaian public university setting, a combination rarely examined in the internal audit

literature.

2.2.2 Demographic influences on technology adoption perceptions

Demographic characteristics have been widely recognised as important moderators of technology adoption behaviour (Chawla & Joshi, 2018; Ibrahim, 2018). Prior studies highlight those factors such as age, gender and education level shape individuals' perceptions, readiness, and behavioural intentions toward new technologies (Naatu et al., 2025). The following subsections review empirical evidence on each of these demographic dimensions.

1. Age

Age has been consistently identified as a significant sociodemographic determinant of attitudes toward AI adoption. In Ghana, Sobiesuo et al. (2025) conducted a cross-sectional survey of 407 residents in the Greater Kumasi area using the TAM and found that sociodemographic factors, including age, education, and gender, significantly influenced PU of AI. Younger residents demonstrated higher AI literacy and more favourable perceptions of AI's reliability, while older respondents expressed greater scepticism, particularly regarding AI decision-making in sensitive domains such as healthcare and law enforcement. Similarly, Nyarko (2024) found that among Ghanaian tertiary students, trust in AI-based health information varied by age and gender, with younger females exhibiting the highest levels of trust. A 2026 study on AI readiness among Ghanaian youth further confirmed that those under 25 years were significantly more likely to use AI tools than those aged 25 and above ($\chi^2 = 4.87, p = 0.03$), attributing this pattern to greater familiarity with emerging digital platforms among younger cohorts. These generational differences are largely mediated by disparities in digital literacy, educational attainment, and access to digital infrastructure, barriers that disproportionately affect older Ghanaians, particularly in rural areas as indicated in the Ghana's National AI Strategy (2023–2033). Globally, the OECD (2025) corroborates these findings, noting that age is a primary driver of generative AI adoption, with individuals aged 18 to 35 being the most optimistic about AI's usefulness and trustworthiness. Taken together, the literature underscores the need for age-sensitive AI policies and inclusive digital literacy programmes to ensure equitable AI adoption across Ghana's diverse population.

2. Gender

Gender has emerged as a significant sociodemographic variable shaping perceptions and adoption of AI, both globally and within the Ghanaian context. Globally, a consistent gender gap in AI adoption has been observed, with men using AI more frequently and expressing more positive attitudes toward it than women (Tang et al., 2025). Research using the TAM and the UTAUT consistently finds that men are more inclined to adopt new information technologies, while women tend to report higher levels of anxiety and ethical concern regarding AI use, which can reduce self-efficacy and slow adoption (Aguirre-Urreta & Marakas, 2010; Tang et al., 2025).

In Ghana specifically, Sobiesuo et al. (2025) found that gender, alongside age and education, significantly influenced PU of AI among residents in the Greater Kumasi area. Nyarko (2024) similarly found gender-differentiated trust in AI-based health information among Ghanaian tertiary students, with young females exhibiting the highest trust levels, a finding that nuances the broader pattern by suggesting that within younger, educated cohorts, women may be more receptive to AI in specific domains such as healthcare.

3. Educational level

Educational level has been widely identified as one of the most critical determinants of AI perception and adoption, particularly in developing country contexts such as Ghana. Baidoo-Anu et al. (2024), in their systematic review of AI acceptance and usage in sub-Saharan African education, found a concentration of AI-related studies originating from Ghana, underscoring the country's centrality in regional AI education research. Their review highlighted that students and staff with higher educational attainment demonstrated significantly greater awareness of AI tools, more positive attitudes toward their integration, and stronger intentions to adopt them, patterns consistent across both tertiary and secondary education levels. This is corroborated by Elliason & Khajuria (2025), whose survey of Ghanaian university academics found that digital literacy, closely associated with educational level, was a key enabler of AI adoption, while limited digital literacy among less formally educated staff posed a significant barrier, particularly in public universities. These findings collectively affirm that educational level functions not merely as a demographic variable but as a proxy for digital literacy, critical thinking capacity, and infrastructure access, all of which shape how Ghanaians perceive and engage with AI technologies.

3. Methodology

3.1 Research Design and Philosophy

This study adopts a quantitative research design situated within a positivist philosophical tradition (Maksimović & Evtimov, 2023). The quantitative methods are appropriate for systematically examining patterns and associations across a defined population using standardized measurement instruments (Rahi, 2017). A cross-sectional survey approach was employed, enabling the collection of data from multiple institutions and respondent

categories at a single point in time (Rindfleisch et al., 2008). This design is consistent with established practice in technology adoption research and is well-suited to the study's objective of describing and explaining demographic variation in AI adoption perceptions.

3.2 Population and Sampling

The study population comprises internal auditors employed across six public universities in Ghana, namely University of Ghana (UG), Kwame Nkrumah University of Science and Technology (KNUST), University of Cape Coast (UCC), University of Education, Winneba (UEW), Ghana Communication Technology University (GCTU), and University of Energy and Natural Resources (UENR). These institutions were selected from the sixteen public universities listed by the Ghana Tertiary Education Commission. These six institutions were intentionally chosen to capture variation in size, funding profile, and internal audit maturity across Ghana's public university system: UG and KNUST represent the largest, most resource-endowed institutions; UCC and UEW represent established mid-sized universities; and GCTU and UENR represent smaller, newer, and more technologically oriented institutions. This spread means that, while not exhaustive of all sixteen public universities, the sample is reasonably reflective of the range of institutional contexts in which internal auditors in Ghana's public university sector operate. The total audit staff numbered 180. Using the formula (Yamane, 1973), 177 respondents were determined as the sample, representing 98% with a 2% margin of error. Roscoe (1975) recommends a minimum of 30 for statistical validity, while Agresti et al. (2017) reaffirm this benchmark. Thus, the 177-sample size ensured both statistical power and representativeness.

3.3 Data Collection Instruments

Data were collected using a structured self-administered questionnaire comprising two sections. Section A captured respondent demographic information: age group, gender and educational qualification as an internal auditor in the public university. Section B measured perceptions of AI adoption using a five-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree). The AI adoption perception construct was operationalised through items adapted from validated TAM and UTAUT scales, covering four dimensions: Fraud Detection Effectiveness, AI Adoption & Accountability, Implementation Challenges and Strategies & Enablers. Items were adapted to reflect the specific context of internal auditing in Ghanaian public universities. To assess the internal consistency of the questionnaire, Cronbach's alpha was computed for the AI adoption perception scale. The overall reliability coefficient was 0.707, indicating acceptable internal consistency and confirming that the instrument was sufficiently reliable for measuring perceptions of AI adoption among internal auditors. Content validity was established through expert review by a professor in accounting and senior practitioners from the IIA of Ghana. A composite AI adoption perception score was computed for each respondent by averaging responses across all perception items, with higher scores indicating more favourable perceptions.

3.4 Data Analysis Strategy

Analyses were conducted using IBM SPSS Statistics (Version 26). The analytical strategy proceeded in two stages. First, descriptive statistics—frequencies, means, and standard deviations—were computed to profile the sample and to describe AI adoption perceptions within each demographic category. Secondly, multiple regression analysis was performed to identify the relative contribution of each demographic variable to the explained variance in AI adoption perceptions, while controlling for the influence of the other variables. Regression assumptions, including normality of residuals, homoscedasticity, and absence of multicollinearity, were verified prior to interpretation of results.

4. Results

4.1 Demographic Profile of Respondents

As can be seen in Table 1, the sample was predominantly male (63.8%, $n = 113$), with female respondents constituting 36.2% ($n = 64$), reflecting the documented gender skew in professional audit staffing across Ghana's public sector—IAA. Most respondents (49.7%) fell within the 31–40 age bracket, indicating a workforce concentrated in mid-career professionals. Younger auditors aged 20–30 represented 19.2%, and those aged 50–60 constituted the smallest cohort (10.7%), consistent with the demographic structure of Ghanaian public institutions. In terms of educational attainment, 64.4% of respondents held postgraduate qualifications, either a master's degree alone (33.3%) or a master's combined with professional qualifications such as Association of Chartered Certified Accountants (ACCA), Chartered Institute of Management Accountants (CIMA), Institute of Chartered Accountants (ICA), or Certified Internal Auditor (CIA) (31.1%). A further 27.7% held a bachelor's degree, while

7.3% held Higher Advanced Diploma (HND).

Table 1. Demographic profile of respondents ($N = 177$)

Category	Sub-Category	<i>n</i>	Percentage (%)
Gender	Male	113	63.8
	Female	64	36.2
Age group	20–30 years	34	19.2
	31–40 years	88	49.7
	41–50 years	36	20.3
	50–60 years	19	10.7
	Higher Advanced Diploma (HND)	13	7.3
Educational qualification	Bachelor's degree	49	27.7
	Master's degree	59	33.3
	Master's + Professional qualifications	55	31.1
	Doctorate + Professional qualifications	1	0.6

Note: N = total sample size; n = subgroup sample size.

4.2 Descriptive Statistics of Artificial Intelligence Adoption Perceptions

As can be seen in Table 2, Prior to examining demographic variations, the overall distribution of AI adoption perceptions was assessed across the four construct dimensions: AI Adoption & Accountability, Fraud Detection Effectiveness, Implementation Challenges, and Strategies & Enablers. A composite score was computed by averaging item responses across all constructs (excluding the Challenges dimension, which was scored in the context of barriers rather than adoption favourability). All mean scores fell within the “Agree” range on the five-point Likert scale ($M = 3.53$ to $M = 3.63$), with relatively low standard deviations ($SD = 0.28$ – 0.37), indicating substantial uniformity in respondent perceptions. The overall composite score was $M = 3.614$ ($SD = 0.287$), confirming a moderately positive orientation toward AI adoption across the surveyed population.

Table 2. Descriptive statistics of artificial intelligence adoption perception constructs

Construct/Dimension	<i>N</i>	Mean	Standard Deviation	Interpretation
AI Adoption & Accountability	177	3.634	0.301	Agree
Fraud detection effectiveness	177	3.591	0.327	Agree
Implementation challenges	177	3.558	0.309	Agree
Strategies & Enablers	177	3.619	0.374	Agree
Overall composite score	177	3.614	0.287	Agree

Note: N = total sample size.

4.3 Artificial Intelligence Adoption Perceptions by Gender

Table 3 presents mean composite and construct-level scores disaggregated by gender. Male respondents recorded a marginally higher overall composite score ($M = 3.629$, $SD = 0.302$) than female respondents ($M = 3.586$, $SD = 0.259$). Prior to interpreting the t-test, Levene's test for equality of variances was conducted and was non-significant, supporting the assumption of equal variances; accordingly, equal variances assumed values are reported. However, an independent samples t-test confirmed that this difference was not statistically significant ($t = 0.954$, $df = 175$, $p = 0.341$). Similarly, no significant gender-based differences were detected across any of the four construct dimensions. Notably, female respondents scored slightly higher than males on the Challenges dimension (Female $M = 3.575$ vs. Male $M = 3.549$), suggesting somewhat greater acknowledgment of implementation barriers, though this difference was also non-significant. These findings are broadly consistent with the UTAUT prediction that gender moderates technology adoption but suggest that within the specific professional context of internal auditing in Ghanaian public universities, gendered differences in AI perceptions may be attenuated by shared institutional socialisation, professional norms, and common exposure to structured audit training environments.

4.4 Artificial Intelligence Adoption Perceptions by Age Group

Table 4 presents mean scores across four age cohorts. A consistent monotonic decline in composite AI adoption perceptions was observed with increasing age: the 20–30 age group recorded the highest mean ($M = 3.659$, $SD = 0.319$), followed by 31–40 ($M = 3.617$, $SD = 0.284$), 41–50 ($M = 3.604$, $SD = 0.304$), and 50–60 ($M = 3.534$, $SD = 0.199$). A one-way ANOVA indicated that these differences did not reach conventional statistical significance for the composite score ($F(3, 173) = 0.780$, $p = 0.507$). However, a supplementary Pearson correlation analysis

revealed a statistically significant negative relationship between age and the Strategies & Enablers dimension specifically ($r = -0.156$, $p = 0.039$), indicating that younger auditors expressed significantly greater endorsement of AI enablement strategies and training interventions than their older counterparts. This pattern aligns with generational digital literacy theory and is consistent with Ghana-specific findings by Sobiesuo et al. (2025) and OECD (2025), which document younger cohorts as more optimistic about AI usefulness and more receptive to AI-related organisational strategies.

Table 3. Artificial intelligence adoption perceptions by gender ($N = 177$)

Construct	Male <i>n</i>	Male Mean	Male Standard Deviation	Female <i>n</i>	Female Mean	Female Standard Deviation	<i>t/p</i>
AI adoption & accountability	113	3.643	0.311	64	3.617	0.284	ns
Fraud detection	113	3.617	0.333	64	3.544	0.313	ns
Challenges	113	3.549	0.318	64	3.575	0.296	ns
Strategies & enablers	113	3.625	0.394	64	3.609	0.337	ns
Overall composite	113	3.629	0.302	64	3.586	0.259	$t = 0.954$ $p = 0.341$

Note: N = total sample size; n = subgroup sample size; t = independent-samples t -test statistic; p = significance level (two-tailed); ns = not statistically significant ($p > 0.05$).

Table 4. Artificial intelligence adoption perceptions by age group

Construct	20–30 Mean	20–30 Standard Deviation	31–40 Mean	31–40 Standard Deviation	41–50 Mean	41–50 Standard Deviation	50–60 Mean	50–60 Standard Deviation	<i>F/p</i>
Accountability	3.700	0.332	3.628	0.286	3.617	0.291	3.528	0.274	ns
Fraud detection	3.659	0.341	3.601	0.316	3.561	0.342	3.491	0.218	ns
Challenges	3.614	0.277	3.559	0.316	3.558	0.317	3.496	0.282	ns
Strategies	3.747	0.434	3.611	0.347	3.575	0.401	3.461	0.328	$p = 0.039^*$
Composite	3.659	0.319	3.617	0.284	3.604	0.304	3.534	0.199	$F = 0.78$ $p = 0.507$

Note: n = subgroup sample size; F = one-way analysis of variance (ANOVA) test statistic; p = significance level; ns = not statistically significant ($p > 0.05$); * = statistically significant at $p < 0.05$.

4.5 Artificial Intelligence Adoption Perceptions by Educational Qualification

Table 5 presents mean composite scores across educational qualification levels. A positive, monotonically increasing pattern was observed, with respondents holding master's degrees combined with professional qualifications recording the highest mean ($M = 3.675$, $SD = 0.335$), compared to respondents with an advanced diploma/HND ($M = 3.548$, $SD = 0.148$) at the lower end. A one-way ANOVA did not yield a statistically significant overall effect ($F(4, 172) = 0.988$, $p = 0.415$). Nevertheless, the regression analysis in Section 4.6 reveals that educational level is a statistically significant independent predictor of composite AI adoption perceptions when controlling for other demographics ($B = +0.064$, $t = 2.713$, $p = 0.007$). This discrepancy between bivariate ANOVA and multivariate regression results reflects suppression effects, where the influence of education on AI perceptions only becomes statistically apparent once collinear effects of age and gender are controlled. The positive perceptions affirms that higher educational attainment, particularly when combined with professional certifications, is associated with more favourable perceptions of AI adoption, consistent with Baidoo-Anu et al. (2024) and Elliason & Khajuria (2025) on the role of educational level as a digital literacy proxy in Ghanaian professional contexts.

Table 5. Artificial intelligence adoption perceptions by educational qualification (composite score)

Construct	HND <i>n</i>	HND Mean	BSc <i>n</i>	BSc Mean	MSc <i>n</i>	MSc Mean	MSc + Prof. <i>n</i>	MSc + Prof. Mean	<i>F/p</i>
Composite score	13	3.548	49	3.582	59	3.597	55	3.675	$F = 0.99$ $p = 0.415$

Note: HND = Higher National Diploma; BSc = Bachelor of Science; MSc = Master of Science; Prof. = Professional; n = subgroup sample size; F = one-way analysis of variance (ANOVA) test statistic; p = significance level. Statistical significance was assessed at the 5% level ($\alpha = 0.05$).

4.6 Multiple Regression Analysis: Demographic Predictors of Artificial Intelligence Adoption Perceptions

As can be seen in Table 6, multiple ordinary least squares (OLS) regressions were conducted with composite AI adoption perception scores as the dependent variable and age, gender and educational level as independent variables. The overall model was statistically significant ($F(3, 173) = 3.098, p = 0.028$), explaining 5.1% of total variance in AI adoption perceptions ($R^2 = 0.051$, Adjusted $R^2 = .035$). While the explained variance is modest, the model's significance attests to a meaningful collective contribution of demographic characteristics to perception variability. Two predictors were individually significant: Age ($B = -0.058, \beta = -0.185, SE = 0.025, t = -2.276, p = 0.024$) and Educational Level ($B = +0.064, \beta = +0.220, SE = 0.024, t = +2.713, p = 0.007$). Educational level was the strongest predictor ($\beta = +0.220$), followed by age ($\beta = -0.185$). The negative coefficient for age indicates that, controlling for other variables, each successive age bracket was associated with a 0.058-point decrease in composite AI adoption perception, confirming that younger auditors held systematically more favourable AI orientations. The positive coefficient for educational level indicates that each upward step in educational attainment was associated with a 0.064-point increase in composite AI perceptions, supporting the notion that education functions as a cognitive and digital literacy facilitator for AI acceptance (Baidoo-Anu et al., 2024). Gender was not a significant independent predictor ($B = -0.018, \beta = -0.032, p = 0.664$). Regression diagnostics confirmed that multicollinearity assumptions were satisfied: VIF values ranged from 1.001 to 1.201 (all well below the conventional threshold of 5.0), tolerance values ranged from 0.833 to 0.999, and the condition number was 20.0, well below the threshold of 30—collectively confirming the absence of multicollinearity. The Durbin-Watson statistic was 1.347, marginally below the conventional lower bound of 1.5, suggesting mild positive autocorrelation in residuals. For cross-sectional survey data this is not an unusual finding and is likely attributable to natural clustering among respondents of similar age and educational background rather than a structural model violation; this is noted as a minor limitation. Inspection of standardised residual plots confirmed homoscedasticity and approximate normality of residuals.

Table 6. Multiple regression results: demographic predictors of composite artificial intelligence adoption perception score

Predictor	<i>B</i>	β	<i>SE</i>	<i>t</i>	<i>p</i>	95% CI	VIF
Constant	+3.566	—	0.093	38.533	< 0.001***	[3.383, 3.749]	—
Age	-0.058	-0.185	0.025	-2.276	0.024*	[-0.107, -0.008]	1.200
Gender	-0.018	-0.032	0.042	-0.435	0.664 (ns)	[-0.102, +0.065]	1.001
Educational level	+0.064	+0.220	0.024	+2.713	0.007**	[+0.017, +0.110]	1.202
Model summary	—	—	—	—	—	<i>N</i> = 177	DW = 1.347
$R^2 = 0.051$	Adj. $R^2 = 0.035$	—	$F(3, 173) = 3.098$	$p = 0.028$	—	—	—

Note: *B* = unstandardised regression coefficient; β = standardised regression coefficient; *SE* = standard error; *t* = *t*-statistic; *p* = significance level; CI = confidence interval; VIF = variance inflation factor; *N* = total sample size; DW = Durbin-Watson statistic; * = statistically significant at $p < 0.05$; ** = statistically significant at $p < 0.01$; *** = statistically significant at $p < 0.001$; ns = not statistically significant ($p > 0.05$); "—" indicates not applicable.

5. Discussion

5.1 Age and Perceptions of Artificial Intelligence Adoption

Age was a statistically significant negative predictor of AI adoption perceptions in the regression model ($B = -0.058, \beta = -0.185, SE = 0.025, p = 0.024$), confirming UTAUT's proposition that older individuals are less influenced by performance expectancy in forming technology adoption intentions (Venkatesh et al., 2003). Descriptively, composite means declined monotonically from the youngest cohort (20–30 years, $M = 3.659$) to the oldest (50–60 years, $M = 3.534$), consistent with the TAM-based findings of Sobiesuo et al. (2025), who reported that younger Ghanaian residents demonstrated higher AI literacy and more favourable AI perceptions. The age effect was most pronounced on the Strategies and Enablers dimension ($r = -0.156, p = 0.039$), indicating that older auditors are significantly less supportive of institutional AI enablement strategies, a pattern aligned with OECD (2025) evidence that individuals aged 18–35 show the strongest support for AI-enabling environments globally. Although the bivariate ANOVA was non-significant ($F = 0.780, p = 0.507$), the regression result confirms that age carries independent explanatory power beyond other demographics, consistent with Ghana's National AI Strategy (2023–2033), which identifies generational digital literacy disparities as a structural challenge to equitable public sector AI adoption.

5.2 Gender and Perceptions of Artificial Intelligence Adoption

Contrary to the dominant global literature, gender was not a significant predictor in either the regression model ($B = -0.018$, $\beta = -0.032$, $p = 0.664$) or bivariate analysis ($t = 0.954$, $p = 0.341$), with negligible differences across all four constructs. This diverges from the global finding of Tang et al. (2025) of a consistent male advantage in AI adoption and from UTAUT's prediction that gender moderates performance expectancy (Venkatesh et al., 2003). However, the null result is coherent with this sample's high educational homogeneity, over 64% of respondents held postgraduate qualifications, which likely suppresses the education-mediated gender gap that drives differences in broader population studies (Baidoo-Anu et al., 2024). In the regression model, education absorbs the gender effect, rendering its independent coefficient non-significant. This aligns with Nyarko (2024), who observed that within educated Ghanaian professional cohorts, gender-based AI perceptual differences are attenuated. The marginal descriptive male advantage nonetheless warrants monitoring, as Tang et al. (2025) note that gender gaps in AI tend to widen as applications become more technically demanding.

5.3 Educational Level and Perceptions of Artificial Intelligence Adoption

Educational level was the strongest predictor in the regression model ($B = +0.064$, $\beta = +0.220$, $SE = 0.024$, $p = 0.007$), confirming that each upward step in educational attainment is associated with more favourable AI adoption perceptions, independent of age and gender. The highest composite mean was recorded by respondents combining postgraduate degrees with professional qualifications such as ACCA, CIMA, ICA, or CIA ($M = 3.675$), compared to Advanced Diploma holders ($M = 3.548$). The one-way ANOVA was non-significant ($F = 0.988$, $p = 0.415$), but the regression finding reveals a real effect masked at the bivariate level by suppressor dynamics, a methodological distinction of substantive importance. This result is consistent with Baidoo-Anu et al. (2024) systematic review documenting that higher educational attainment is the most consistent enabler of positive AI perceptions in sub-Saharan Africa, and with Elliason & Khajuria (2025) finding that digital literacy, closely associated with education, is the primary structural enabler of AI adoption among Ghanaian university staff. Within TAM, the finding affirms that education builds the PU and PEOU cognitions that Davis (1989) identified as the core antecedents of technology adoption. The compounded effect of academic and professional credentialing in the highest-scoring group further reflects UTAUT's treatment of "experience" as a positive moderator of effort expectancy (Venkatesh et al., 2003).

5.4 Implications for Routine Internal Audit Activities

Beyond the technology-adoption implications in general, the findings also matter for everyday internal audit work. Since younger and educated auditors tend to feel more comfortable with AI, audit teams should keep this in mind when deciding who works on AI-based tasks, pairing auditors who are less sure about AI with colleagues who are more confident can help the rollout or usage to be smooth. When it comes to risk identification, auditors overall showed a positive attitude toward AI, which means audit directorates, departments or units are in a good position to start using AI tools to flag risks and unusual activity, as long as staff who are less familiar with these tools get proper training. AI tools also only work well if people keep using them regularly, not just once, so knowing that age and education affect comfort with AI helps teams figure out who will need extra support to keep using these tools consistently. Finally, since older auditors and those with less formal education feel less confident with AI, training budgets should focus more on this group, so that AI tools end up being used across the whole team rather than just by a small group of early adopters.

6. Conclusion

This study examined how age, gender, and educational level shape internal auditors' perceptions of AI adoption across six public universities in Ghana ($N = 177$). Anchored in TAM and UTAUT the findings confirm that internal auditors hold broadly positive AI adoption perceptions (composite $M = 3.614$). The overall regression model was significant ($F(3, 173) = 3.098$, $p = 0.028$, $R^2 = .051$), with age ($B = -0.058$, $\beta = -0.185$, $p = 0.024$) and educational level ($B = +0.064$, $\beta = +0.220$, $p = 0.007$) emerging as the two independent demographic predictors. Gender was non-significant ($p = 0.664$), explained by the sample's educational homogeneity suppressing the education-mediated gender effect. These findings extend TAM and UTAUT validation to the underexplored context of professional internal audit in Sub-Saharan Africa and confirm that age-sensitive and education-focused workforce interventions are the most empirically justified demographic priorities for advancing AI adoption in Ghana's public university audit sector.

7. Recommendations

Age-differentiated AI training

The IAA and university management should design AI training curricula adjusted to different age groups rather than uniform programmes, given the significant age effect on AI adoption perceptions, particularly on institutional enablement strategies ($r = -0.156$, $p = 0.039$). Foundational modules should target older staff to build digital confidence, while advanced modules for younger auditors should focus on critical evaluation of AI outputs and ethical considerations.

Education and professional qualification upgrade

Since educational level is the strongest regression predictor ($B = +0.064$, $\beta = +0.220$, $p = 0.007$), the IAA and Ghana Tertiary Education Commission should formalise education upgrade pathways for audit staff, including cost-sharing sponsorship for postgraduate study. The IIA of Ghana, the ICA, Ghana, and the Association of Chartered Accountants should integrate AI competency modules into continuous professional development renewal requirements, reinforcing the combined academic-professional qualification pathway that produced the highest AI adoption perceptions in this study.

Staff development and workforce planning

Since age and education level are both closely linked to how auditors feel about AI, audit directorates, departments or units should factor these into their staff development plans instead of giving everyone the same training. This could mean pairing younger, tech-savvy auditors with senior colleagues on AI-related work so that both can learn from each other; offering different levels of training, with basic digital skills courses for staff without postgraduate qualifications and more advanced AI and analytics courses for those with postgraduate or professional qualifications; and rolling out AI-related changes to job roles gradually over several years, so that older or less formally qualified staff aren't overwhelmed all at once. Building these steps into yearly staff reviews and succession planning would help audit directorates, departments or units turn these findings into real, practical actions rather than just general policy statements.

National AI audit framework

Given that demographics explain only 5.1% of variance ($R^2 = 0.051$), organisational and policy-level factors account for the larger share of AI adoption variation. The IAA should develop a Ghana Public Sector Internal Audit AI Adoption Framework specifying minimum competency standards, approved tools, data governance requirements, and ethical guidelines, operationalising Ghana's National AI Strategy (2023–2033) at the institutional audit level.

8. Limitations and Future Research

This study only looked at six public universities in Ghana, so the results may not apply to other public sector organizations or private companies. Private organizations are different from public universities in things like funding, oversight rules, and workplace culture, and these differences could affect how they adopt AI in ways this study didn't capture. For example, private companies might have more freedom to buy AI tools without going through lengthy approval or procurement processes, but fewer legal requirements are pushing them to use those tools. So, the patterns found in this study, based on public university staff, should be seen as a general indication rather than something that applies directly to private companies. The regression model explains 5.1% of variance, indicating that unmeasured organisational and individual factors account for most of the perceptual variation. The regression model explains 5.1% of variance, indicating that unmeasured organisational and individual factors account for most of the perceptual variation. The Durbin-Watson statistic of 1.347 indicates mild positive autocorrelation in residuals, likely reflecting natural respondent clustering by age and educational background in a cross-sectional sample rather than a structural model violation.

Future research should address these limitations through:

1. Longitudinal studies tracking how age and education affect AI adoption perceptions evolve as AI tools become more embedded in audit practice.
2. Qualitative inquiry to unpack the mechanisms behind the regression-identified age and education effects.
3. Extension to other Ghanaian public sector domains and comparative studies across Sub-Saharan Africa using the same instrumentation.
4. Examination of whether positive AI adoption perceptions translate into measurable audit quality improvements such as higher fraud detection rates and expanded audit coverage.

Author Contributions

Conceptualization, J.C.W.; methodology, J.C.W.; formal analysis, J.C.W.; investigation, J.C.W.; data curation, J.C.W.; writing—original draft preparation, J.C.W.; writing—review and editing, J.C.W., M.D., and K.B.; visualization, J.C.W.; supervision, M.D. and K.B.; validation, M.D. and K.B. All authors have read and agreed to

the published version of the manuscript.

Ethical Approval

Ethical clearance approval with code HSSREC/00008064/2024 was obtained from the University of KwaZulu-Natal ethics board prior to data collection. Participants were informed about the purpose of the study, their right to withdraw at any time, and the confidentiality of their responses. Informed consent was obtained in both written and verbal forms before the administration of questionnaires.

Data Availability

The data supporting this study's findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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